Decomposition method

Stepwise replacement

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Learning objective

By the end of this lecture, you will understand

1. what the stepwise replacement method for decomposition is
2. how to apply this method in R
3. the adv/disadv when we compare this method and Horiuchi method

Recap

What did we learn yesterday?

Decomposition methods – Horiuchi method, and a recap of the fundamental calculus.

Goal of decomposition: break down the difference in a total metric into the contributions of its underlying covariates;

and answer: which covariate matters most?

Decomposition

An indicator y is a function of n demographic variables.

$$y = f(x_1, x_2, \ldots, x_n)$$

Can we think about the Game system (players)?

Examples:

life expectancy as a function of age-specific mortality;

crude death rate as a function of mortality rates and population structure.

Decomposition

$$y = f(x_1, x_2, \dots, x_n)$$

Two population states are observed: $x^{(1)} \rightarrow x^{(2)}$

Total change: $\Delta y = f\left(x^{(2)}\right) - f\left(x^{(1)}\right) = y_2 - y_1$

$$\Delta y = y_2 - y_1 = \sum_i c_i$$

Question: How much does the change in each covariate contribute to the overall change in the function?

Motivation to stepwise replacement

We have learnt the linear integral model (Horiuchi method).

What if

- The function is not smooth and not differentiable?
- Complicated indicators?
- Deriving the partial derivative analytically is difficult or intractable?

The stepwise replacement method

Now, let's study a simpler method ---

Stepwise replacement method:

Approximate the continuous integral using sequential discrete replacements of variables.

Adv: simplicity and transparency; no need for the derivatives.

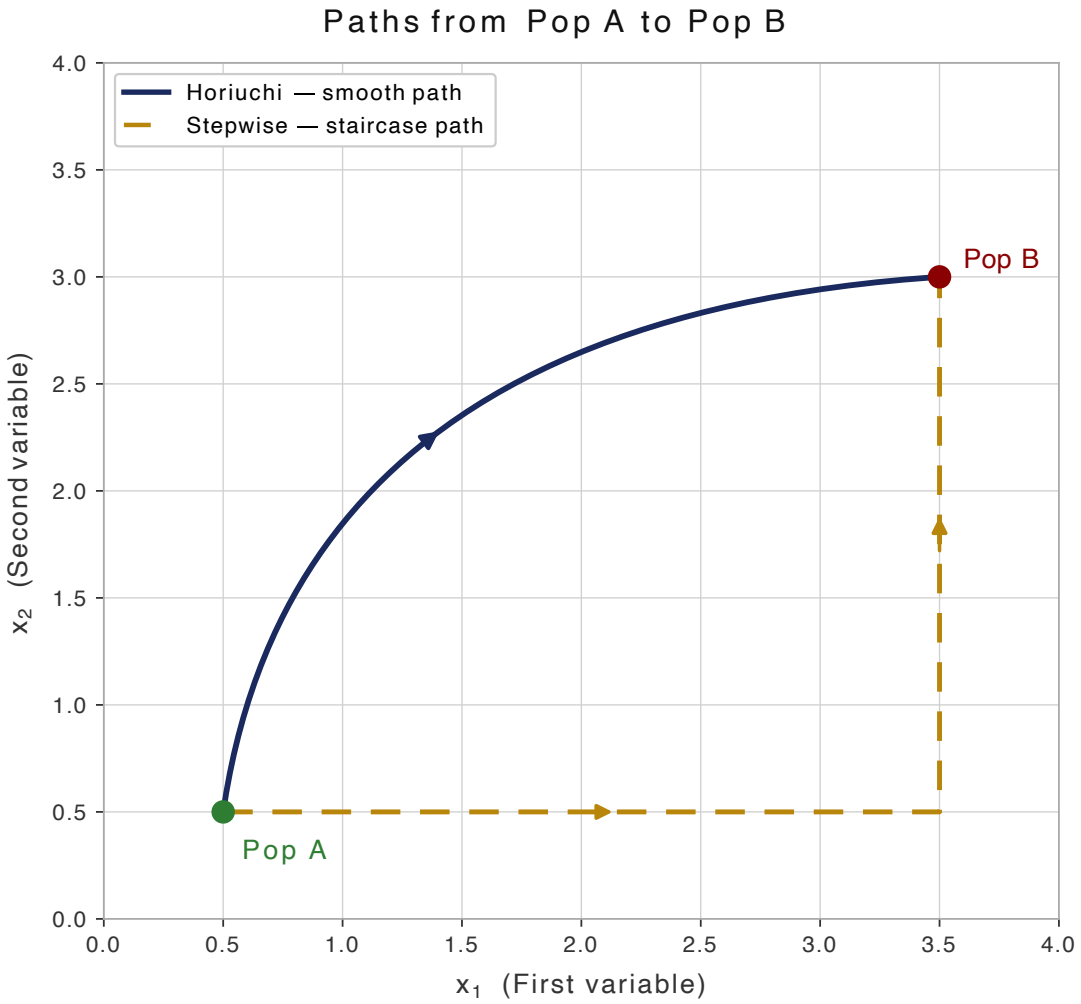
Decomposition methods comparison

- Horiuchi:

Allow all ages to change simultaneously, integrating contributions along a smooth continuous path.

- Stepwise replacement:

Change one variable at a time, observing the incremental effect before proceeding to the next variable .



Algorithm

MIRG
Mortality and Inequalities Research Group

Stepwise Decomposition of Life Expectancy Differences

Attributing the gap in e_0 between two populations to age-specific mortality contributions

Population A mortality rate m_x^A | Population B mortality rate m_x^B | Age being substituted in current step | Resulting age-specific contribution Cx

0 · Setting

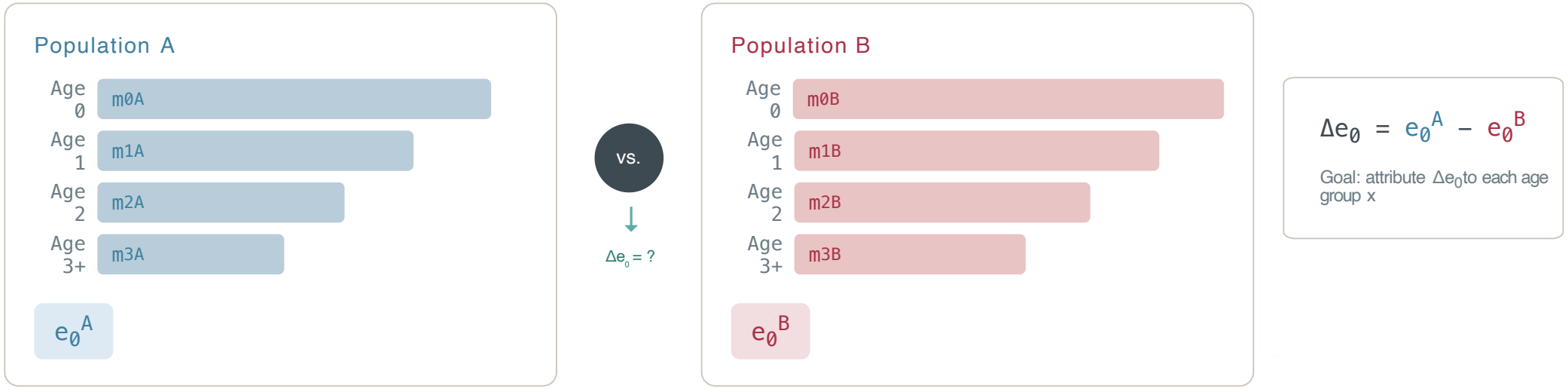


Figure layout and code were initially generated by Claude Sonnet 4.6 (Anthropic, 2026) based on author-specified algorithmic content, then manually revised by the author.

Algorithm

1 · Stepwise replacement

Step 1 · Age 0

Pop B → substitute m_0^A

POP A	POP B*
m0A	m0A✦
m1A	m1B
m2A	m2B
m3A	m3B

e_0^B → $e_0^{B'}$

CONTRIBUTION

$$C_0 = e_0^{B'} - e_0^B$$

Replace age-0 mortality of Pop B with Pop A value; compute new e_0

Step 2 · Age 1

Pop B′ → substitute m_1^A

POP A	POP B′′
m0A	m0A
m1A	m1A✦
m2A	m2B
m3A	m3B

$e_0^{B'}$ → $e_0^{B''}$

CONTRIBUTION

$$C_1 = e_0^{B''} - e_0^{B'}$$

Using the updated life table, substitute age-1 mortality; compute incremental change

Step 3 · Age 2

Pop B′′ → substitute m_2^A

POP A	POP B′′′
m0A	m0A
m1A	m1A
m2A	m2A✦
m3A	m3B

$e_0^{B''}$ → $e_0^{B'''}$

CONTRIBUTION

$$C_2 = e_0^{B'''} - e_0^{B''}$$

Substitute age-2 mortality in the evolving hybrid life table

Step 4 · Age 3+

Pop B′′′ → substitute m_3^A

POP A	POP B′′′′
m0A	m0A
m1A	m1A
m2A	m2A
m3A	m3A✦

$e_0^{B'''}$ → e_0^A

CONTRIBUTION

$$C_3 = e_0^A - e_0^{B'''}$$

Final substitution restores the fully Pop-A life table, yielding e_0^A

2 · Summation and sanity check

$$C_0 + C_1 + C_2 + C_3 + \ldots = \Sigma C_x = \Delta e_0$$

HTML in GitHub
https://github.com/YuCHENJT/PDP_2025-2026_lecture_materials/tree/main/Decomposition%20method/Lecture%20notes

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R lab

Let's use Stepwise replacement decomposition in R

DemoDecomp::stepwise_replacement

We are going to compare Horiuchi method and Stepwise replacement method.

Does ordering matter?

Function used for decomposition:

Non-linear (includes interaction terms)? Correlation across components? – think about the partial derivatives.

An example in HTML in GitHub

Advanced decomposition method:

- Shapley value function

Method comparison

Feature	Horiuchi Method	Stepwise Replacement
Type	Continuous decomposition	Discrete decomposition
Mathematical basis	Calculus (integration)	Finite differences
Interpretation	Marginal effects integrated along path	Sequential variable replacement
Path dependence	Minimal	Possible (order matters)
Implementation	More complex	Easy
Flexibility	Requires differentiable function	Works for almost any function

Which method to use?

Horiuchi Method	Stepwise Replacement
Theoretical mathematical clarity and accurate are required.	The indicator function is highly complex or analytically opaque.
The demographic function is smooth and differentiable.	Partial derivatives are difficult or impossible to compute.
A continuous decomposition across many interacting variables is needed	A computationally straightforward implementation is required.
The analysis is intended for publication in high-level academic outlets.	Applied demographic work demands flexibility across function types.

Advanced materials

Reading list for your interests

Stepwise Replacement

Andreev, E. M., Shkolnikov, V. M., & Begun, A. Z. (2002). Algorithm for decomposition of differences between aggregate demographic measures and its application to life expectancies, healthy life expectancies, parity-progression ratios and total fertility rates. *Demographic research*, *7*, 499-522.

Gupta, Prithwis Das. "Decomposition of the difference between two rates and its consistency when more than two populations are involved." *Mathematical Population Studies* 3.2 (1991): 105-125.

Advanced materials

Reading list for your interests

Shapley value function – Game theory

Shapley, L. (1953) A Value for n-Person Games. In: Kuhn, H. and Tucker, A., Eds., Contributions to the Theory of Games II, Princeton University Press, Princeton, 307-317. https://doi.org/10.1515/9781400881970-018

Shorrocks, Anthony F. "Decomposition procedures for distributional analysis: a unified framework based on the Shapley value." Journal of Economic Inequality 11.1 (2013): 99-126.

Q & A

Reading for the next lecture

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Lecture notes and R labs:
https://github.com/YuCHENJT/PDP_2025-2026_lecture_materials